



Norma

Carbon dioxide removal prepurchase application 2025

General Application

(The General Application applies to everyone; all applicants should complete this)

Public section

The content in this section (answers to questions 1(a) - (d)) will be made public on the [Frontier GitHub repository](#) after we announce purchases in the summer and fall. Include as much detail as possible but omit sensitive and proprietary information.

Company or organization name

Norma

Company or organization location (we welcome applicants from anywhere in the world)

Paris, France

Name(s) of primary point(s) of contact for this application

Monica Larrazabal, Silvia Pugliese

Brief company or organization description <20 words

Norma is pioneering the only DAC technology that captures CO₂ and recovers >95% of the energy used

1. Public summary of proposed project¹ to Frontier

- a. **Description of the CDR approach:** Describe how the proposed technology removes CO₂ from the atmosphere, including how the carbon is stored for > 1,000 years. Tell us why your system is best-in-class, and how you're differentiated from any other organization working on a similar approach. If your

¹ We use "project" throughout this template, but the term is not intended to denote a single facility. The "project" being proposed to Frontier could include multiple facilities/locations or potentially all the CDR activities of your company.

project addresses any of the priority innovation areas identified in the RFP, tell us how. Please include figures and system schematics and be specific, but concise. 1000-1500 words

Norma’s technology & process

At Norma, we are building the only DAC system that doesn’t spend energy—it borrows it.

Unlike other DAC technologies that consume large amounts of energy to release CO₂ from capture materials in the regeneration process, our electrochemical system is designed to give most of that energy back. Norma’s core innovation allows us to recover >95% of the energy used for CO₂ regeneration—radically reducing net energy use, and costs.

Norma’s process combines advanced energy storage technology with a proprietary design into a single, modular platform.

Step 1: Upstream

Ambient air is pushed through the system. As it passes through, CO₂ is selectively absorbed into the electrolyte and reacts to form bicarbonates.

Step 2: Core cell: CO₂ Capture & Regeneration

Inside the core cell, our electrodes design enables the separation of bicarbonate ions from the electrolyte. This step requires an input of energy. Next, our system releases high-purity CO₂ gas from the bicarbonates and allows us to recover the input energy—ready to fuel the next cycle.

Step 3: CO₂ Dehydration & Compression

The pure CO₂ released from the core cell is then dehydrated and compressed. After this process, the storage partner executes the sequestration

1 Commercial energy storage devices can reach >95% energy efficiency

Figure A: Description of Norma’s 3-step Direct Air Capture process

The CO₂ captured through Norma’s process can be permanently sequestered via storage partners. For this pre-purchase application, Norma proposes to work with C-Questa as their storage partner,

leveraging their second site in France where we propose to co-locate our project. C-Questra have signed a Letter of Support for Norma and have collaborated with our team on project scoping.

C-Questra is a carbon storage company specializing in the full CO₂ storage value chain, from emission sources to sequestration sites. They are developing the EU's first onshore Direct Air Capture and Storage project in France, aiming to remove 100,000 tons of CO₂ annually by 2030, with plans to scale up to megaton levels by 2035. Their approach eliminates the need for expensive midstream and offshore infrastructure, offering a cost-effective and scalable solution for atmospheric CO₂ removal. Their injection methodology guarantees >1,000 years permanence for the stored CO₂ and follows European CCS Directive (Directive 2009/31/EC) for Geological Injection

Norma's Value Proposition & Differentiation

To the best of our knowledge, Norma is the first to commercialize a technology that integrates energy storage devices with CO₂ capture—unlocking an entirely new DAC approach. Norma's technology targets an energy consumption of 250–500 kWh/tCO₂, a tenfold improvement over first generation DAC systems, with a clear pathway to reaching costs well below \$100 per ton.

Norma's new technology features the following advantages:

- **Energy efficiency:** The energy storage component of our tech enables us to overcome the energy required in the solvent regeneration step. This means that for the end to end process, we can target energy demands of 250-500 kWh/tCO₂.
- **Unlocking Grid Synergies Through Energy Storage:** our technology opens up multiple avenues for synergies with grid services. In addition to capturing pure CO₂, the technology has the potential to store or even supply energy, enabling dual-use applications in grid stabilization. This flexibility paves the way for integration into broader energy systems and adaptation to other use cases beyond carbon capture.
- **Minimal Environmental Footprint:** Our process operates at ambient pressure and temperature, eliminating the need for high-energy inputs or specialized infrastructure. We use no hazardous chemicals and generate no toxic waste streams, ensuring clean and safe operations. Additionally, we use cheap, nontoxic, abundant materials with a long lifespan and high recyclability.
- **Engineered for Simplicity & Scale:** Our technology captures pure CO₂ through a highly modular stack architecture, purpose-built to reduce complexity, streamline mass transport, and minimize energy consumption. By prioritizing simplicity in both form and function, we enable efficient, scalable deployment—from the pilot system to full-scale industrial applications.
- **No Complex Supply Chain Dependencies:** unlike systems that require the development of specialized supply chains, our technology builds on well-established, widely available energy storage components—similar to those used in regenerative braking systems within the automotive industry. By leveraging existing industrial infrastructure, we significantly reduce both CAPEX and OPEX, accelerate deployment timelines, and benefit from strong learning curves driven by mature manufacturing ecosystems.
- **Geographically Flexible:** our process maintains high efficiency regardless of local climate conditions—it is not sensitive to relative humidity or temperature. This makes it uniquely adaptable for deployment across a wide range of geographic locations, from arid deserts to humid coastal regions.

Norma is also proud to be the first DAC company co-founded entirely by women, combining complementary expertise in carbon removal, electrochemistry, greenhouse gas management, repeat

startup founding, and business strategy. Since the beginning, we've been grateful to be embedded in a strong and collaborative European network, which continues to support and strengthen our development.

Turning DAC into a Grid Asset

Importantly, our innovation aligns directly with one of the **priority areas outlined in the RFP: "DAC: Efficient and flexible energy use."** Norma's system is designed to operate as a flexible, dispatchable load, unlocking two key value streams that enhance both energy efficiency and economic viability.

First, we can serve as a flexible load for congestion relief, operating intermittently to absorb excess renewable energy during periods of overgeneration. This allows us to participate in grid incentive programs and earn compensation for helping to stabilize the grid, as well as providing access to low electricity prices.

Second, thanks to our embedded energy storage system, we can participate in Balancing Services, particularly in Frequency Restoration Reserve (FRR) markets. Our system can respond rapidly to power needs, helping to maintain grid frequency through near-instantaneous load adjustment.

- b. **Project objectives:** What are you trying to build? Discuss location(s) and scale. What is the current cost breakdown, and what needs to happen for your CDR solution to approach Frontier's cost and scale criteria?² What is your approach to quantifying the carbon removed? Please include figures and system schematics and be specific, but concise. 1000-1500 words

Project Objectives: location and scale

For our Frontier proposal, we are focused on delivering a commercial-scale demonstration plant by 2027. This Demo unit is designed with a capture nameplate capacity of 250 tons of CO₂ per year, representing a critical step in validating the performance, scalability, and reliability of our technology in real-world conditions. The system will operate with an anticipated energy consumption of [~1700kWh/tCO₂, leveraging our energy-efficient electrochemical design (Capture & Regeneration step in the core cell is estimated only at \$300kWh/tCO₂, the rest is energy for upstream & downstream).

The Demo will be located in France, where Norma is headquartered. France offers a uniquely supportive environment for our goals, including:

- Access to low-carbon electricity, essential for minimizing the lifecycle emissions of CDR

² We're looking for approaches that can reach climate-relevant scale (about 0.5 Gt CDR/year at \$100/ton). We will consider approaches that don't quite meet this bar if they perform well against our other criteria, can enable the removal of hundreds of millions of tons, are otherwise compelling enough to be part of the global portfolio of climate solutions.

- technologies;
- Proximity to emerging CO₂ storage projects;
- Availability of early-stage funding mechanisms and innovation support programs.

Our proposed storage partner C-questra is planning a first CO₂ injection site that is expected to be located near Paris, coming live in 2026. We are in conversations to potentially partner to colocate for the second site with our Demo in 2027. This close proximity will not only simplify logistics and reduce transport emissions, but also allow for real-time process monitoring, tighter integration, and enhanced operational oversight, which is essential at these stages of development.

Looking further ahead, our long-term deployment strategy is global but focused. We aim to scale in regions that meet three essential criteria: low-cost, low-carbon electricity—particularly in areas with grid stability or intermittency challenges where our flexible energy use can support grid resilience; Proximity to CO₂ storage or utilization infrastructure for seamless integration; Supportive regulatory and policy frameworks that enable fast, responsible growth.

Current cost breakdown and optimization

Our current system is pre-commercial, at TRL 4, currently capturing gram-scale CO₂ with an energy demand of 4700 kWh/tCO₂. At scale, our TEA projects that our NOAK cost of carbon removal will be \$100-\$160//tCO₂, in line with Frontier's long-term affordability targets. This figure assumes a cost split of approximately 18% CapEx and 82% OpEx. The dominant component of our OpEx is electricity at 38\$/tCO₂, as our DAC process requires 382kWh/tCO₂ (target 250-500 kWh/tCO₂), at a reference electricity price of \$0.10/kWh.

The Demo (250tCO₂/year capacity) will have a Total Capital Cost below \$1Mn (assuming no cost of capital since it's a pre-commercial deployment financed via Venture Capital). The Levelized Cost of Carbon Removal is \$1478/tCO₂, with 48% cost in CapEx and 52% in OpEx.

Norma is focusing on several key drivers to reduce the current cost of CO₂ capture, with a clear roadmap to improve efficiency at every stage of our technology's lifecycle. These drivers are anchored in our ongoing optimization of Key Performance Indicators (KPIs) and are fundamental to achieving our long-term cost targets. The key areas of focus include:

1. Enhancing Specific Capture Rate

By refining our charge-discharge cycling protocols, we are improving the specific capture rate (measured as mol CO₂ per kg per hour). Through iterative optimization, we aim to increase the capture efficiency, thereby lowering the overall energy demand for each ton of CO₂ removed.

2. Improving Roundtrip Energy Efficiency

We are actively working towards achieving greater than 95% roundtrip energy recovery efficiency in our modular energy storage system. This target is within reach through continued innovation in our core-cell design, material selection, and operational conditions. By minimizing energy losses at every stage of the process, we can drastically reduce the operational expenses associated with electricity use, which is a major component of the overall cost.

3. Scaling Electrochemical Cell to Demo-Relevant Airflow

As part of our Demo development, we will scale our electrochemical cell to handle ambient

airflow at larger volumes. This includes integrating appropriate air-handling components (e.g., fan systems) and optimizing system configuration to minimize pressure drop and maintain high CO₂ uptake. This step is critical to validate performance under conditions representative of field deployment.

4. **Scaling Manufacturing Processes**

Scaling the production of our modular stacks is key to reducing costs. By utilizing modular, standardized components, we avoid the need for supply-constrained, high-cost materials that are typically required in more complex systems. Our Demo project will serve as the first small-scale run of our full stack assembly process, providing insights on time, cost, and quality control.

5. **Leveraging Learning Curves and Scale Economies**

We capitalize on economies of scale in both material sourcing and system integration. As we expand our operations and technology beyond FOAK, we benefit from learning curves—gaining efficiencies and insights from each system deployment. The Demo system will allow us to validate repeatable manufacturing steps, assess supplier readiness, and generate real-world cost and performance data—laying the groundwork for more efficient scale-up to FOAK and beyond.

Quantification of CDR

Norma's DAC technology ensures precise monitoring of CO₂ capture and energy consumption, which are critical to maintaining system efficiency and accountability. Our framework integrates high-accuracy sensors within the airflow of the capture system to directly measure CO₂ concentrations in the captured stream, providing continuous real-time data on CO₂ levels and enabling precise tracking of captured amounts. Additionally, we monitor electricity consumption and energy storage within our internal energy system, offering granular insights into the life-cycle emissions, with the necessary frequency for accurate reporting.

To ensure the accuracy, transparency, and integrity of our carbon removal efforts, we will adhere to well-established MRV methodologies, such as Puro's Geologically Stored Carbon methodology (2024) and the Isometric Protocol for Direct Air Capture (12/2023). These protocols allow us to align with customer/regulatory requirements and industry best practices. Norma will implement real-time data reporting to meet the evolving standards for DAC verification.

In collaboration with our CO₂ storage partners, we will collect end-to-end life-cycle emissions data, including emissions related to CO₂ transport and storage. C-questra can allow efficient and durable CO₂ storage and real-time monitoring of sequestration performance using real-time fiber-optic pressure and temperature monitoring, seismic alarms, and history-matched flow simulations to track plume behavior and confirm permanent stratigraphic and solubility trapping. While we have only conducted an internal lifecycle assessment to date focused on emission quantification, we plan to complete a third-party verified LCA before our Demo deployment to meet environmental impact standards and proactively work on public engagement and social license..

Vision and Path to Scaling:

At Norma, our long-term vision is to create a scalable, flexible, and highly efficient CO₂ removal solution that will become a global standard for Direct Air Capture. By leveraging energy storage, we aim to redefine how DAC can function in a dynamic energy market. Our system's ability to store and

release energy flexibly enables us to respond to fluctuating electricity availability, driving down costs and increasing efficiency.

Scaling in the **short to mid-term**, our approach focuses on owning and operating initial facilities. These pilot and FOAK plants will serve as critical learning platforms, where we will fine-tune our technology, optimize operations, and gather valuable engineering insights. By directly operating these facilities, we gain a first-hand understanding of system performance under real-world conditions. This experience is invaluable for refining the operational aspects of our DAC systems and for demonstrating the commercial viability of our technology to investors, regulators, and stakeholders.

As we progress towards **NOAK in the long-term**, our strategy will evolve toward partnering with project developers and scaling through licensing agreements. This transition will allow us to rapidly expand our impact, leveraging external expertise in energy infrastructure while focusing on advancing our core technology and IP. Our **licensing strategy** will be based on a robust IP portfolio, which is a major priority for us at this stage to ensure that we retain control over the unique aspects of our technology.

- c. **Risks:** What are the biggest risks and how will you mitigate those? Include technical, project execution, measurement, reporting and verification (MRV), ecosystem, financial, and any other risks. 500-1000 words

The development of Norma's technology presents several key challenges and risks in the short term. These primarily relate to technological development and engineering, scaling, partnerships and permitting, as well as the dynamic nature of the carbon market landscape.

R&D and engineering

Electrode performance. The adsorption capacity of CO₂ is influenced by variables such as temperature, pressure, and surface chemistry. As this is a novel application, performance optimization is a key part of our ongoing development process. We are conducting a detailed analysis of the trade-offs between these parameters to identify the optimal operating conditions for performance and efficiency.

We have designed our system to accelerate the natural conversion of atmospheric CO₂ into bicarbonate ions, a typically slow process limited by CO₂ solubility at neutral pH. To overcome this, we've optimized the regeneration step to maximize air-liquid contact and minimize diffusion distances. This also ensures efficient CO₂ uptake even under low concentration conditions, supporting consistent and effective operation without requiring excessive airflow.

The energy demand of any DAC technology is inherently high due to the low concentration of CO₂ in the atmosphere. Norma addresses this challenge by integrating energy storage solutions to recover the energy spent during the process. This introduces the added complexity of hardware and methods (for example optimizing charge and discharge cycles) for energy storage systems. To tackle this, Norma is collaborating with experts, advisors, and team members within the ecosystem to gain insights and learn from their expertise.

Scaling

Scaling our technology from prototype to pilot, and ultimately to a demo plant, presents important challenges. System integration and scale-up are well-known risks when deploying a first-of-a-kind system. To address this, we are developing modular units that can be progressively tested, validated, and refined at each stage before full-scale deployment. Our scale-up strategy is incremental by design—we avoid large leaps in scale for each unit. This gradual approach allows us to learn, adapt, and de-risk effectively, without committing excessive capital upfront. It ensures that technical or design issues are identified and resolved early, keeping the overall development process agile, cost-effective, and informed by real operational data.

Achieving FOAK: Norma's core technology leverages well understood principles proven at scale in established applications including energy storage.. However, adapting these principles to DAC and scaling them to industrial-level plants introduces a new set of challenges. To address these risks, we are going to collaborate with experienced EPC companies to ensure successful scale-up. We are already in discussions with experts in both energy storage and CO₂ capture to guide us through the complexities of scaling and optimizing these technologies for commercial deployment.

Partnerships & Permitting

For CO₂ sequestration, we are planning to colocate with C-Questra (geological injection) for their second location in France. This depends on its injection Exploration Permit approval—an external dependency. To mitigate this risk and ensure the best partnership conditions, we will evaluate alternative storage options in parallel and follow closely the progress of C-Questra on their first deployment scheduled for 2026 in Grandpuits, France.

Our collaboration wouldn't represent a first-of-its-kind initiative for C-questra; however, in the event of any unforeseen issues with the mineralization process, the location of our plant in France offers significant advantages for risk mitigation.

Market & Ecosystem Landscape

Beyond technical and operational risks, the voluntary carbon market is evolving, with shifting political and regulatory dynamics. The election of President Trump in the U.S. could weaken policy support for CDR, potentially slowing market growth and investor confidence. While CDR policy momentum in Europe (e.g., EU CRCF, Green Claims, National Strategies) is positive, it is unlikely to accelerate our deployments in the next 2-3 years. Early-stage carbon removal companies must therefore build resilience in uncertain policy environments. To mitigate this, we are diversifying funding sources (including non-dilutive grants), and securing early customers like Frontier would be catalytic in demonstrating market confidence. Importantly, our technology also enables participation in grid service markets - functioning as a flexible load for congestion relief during periods of overgeneration and providing rapid-response balancing services through frequency reserve mechanisms. These capabilities open new revenue streams and position Norma as a grid-interactive DAC solution, especially valuable in renewable-heavy systems. While our core focus remains durable carbon removal, these strategies will accelerate development, improve scalability, and build financial resilience in a dynamic market.

- d. **Proposed offer to Frontier:** Please list proposed CDR volume, delivery timeline and price below. If you are selected for a Frontier prepurchase, this table will form the basis of contract discussions.

Proposed CDR over the project lifetime (tons) <i>(should be net volume after taking into account the uncertainty discount proposed in 5c)</i>	258 tons
Delivery window <i>(at what point should Frontier consider your contract complete? Should match 2f)</i>	2029
Levelized cost (\$/ton CO ₂) <i>(This is the cost per ton for the project tonnage described above, and should match 6d)</i>	\$1478
Levelized price (\$/ton CO ₂) ³ <i>(This is the price per ton of your offer to us for the tonnage described above)</i>	\$1750

³ This does not need to exactly match the cost calculated for "This Project" in the TEA spreadsheet (e.g., it's expected to include a margin and reflect reductions from co-product revenue if applicable).